COMPLETE SOLUTION CLASSIFICATION OF THE $(A_{12}, 1, A_{12})$ PROJECTION FIBER FOR BRAID AND RIGHT-ANGLED ARTIN GROUPS

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with Kirby Problem 2.27 on embeddings and homomorphisms between braid groups and right-angled Artin groups (RAAGs), we classify all solutions in the projection fiber over the pure-braid triple $(A_{12}, 1, A_{12})$: combining an invertible generator transformation to a path RAAG, faithful Artin action, semidirect-product decomposition, and the Crisp–Wiest square-root uniqueness theorem in pure braid groups, every solution in the fiber has the exact form $(u, v, u\alpha(v))$ with $u\alpha(v) = vu$.

1. Main Fiber Classification Theorem

Theorem 1.1. *In the semidirect-product fiber over the pure-braid projection triple $(A_{12}, 1, A_{12})$, an invertible change of generators identifies the fiber kernel with a path right-angled Artin group, and the Crisp–Wiest unique-root property forces every commuting/braid-compatible lift to take the form $(u, v, u\alpha(v))$ satisfying the twisted commutation relation $u\alpha(v) = vu$.*

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